SanTam Net Alert

AN INTELLIGENT EMAIL TO SMS NOTIFICATION SYSTEM

25EEC204 – MINI PROJECT I

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in partial fulfilment for the award of the degree

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BONAFIDE CERTIFICATE

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ABSTRACT

SanTam NetAlert is a full-stack intelligent notification system designed to bridge the communication gap between email-based alerts and real-time mobile SMS notifications. In today's fast-paced academic and professional environment, critical email notifications often go unnoticed due to poor internet connectivity, device silent mode, or email overload. This project addresses this challenge by continuously monitoring a configured Gmail inbox using the IMAP protocol and automatically forwarding important emails as SMS messages to a mobile phone — even without internet access on the receiving device.

The system employs a keyword-based intelligent filtering engine that scans incoming email subjects and body content for user-defined trigger words such as URGENT, Assignment, Deadline, Exam, and Result. Only emails matching these configured keywords trigger an SMS alert, ensuring that users receive meaningful notifications without noise or spam. SMS delivery is achieved through the httpSMS open-source Android gateway, which uses the user's own Android phone as an SMS transmitter via its SIM card, completely eliminating the need for expensive third-party SMS services.

The backend of the system is built using Node.js with Express.js, providing a RESTful API architecture with endpoints for email management, SMS triggering, delivery log retrieval, and dynamic settings configuration. The frontend dashboard is developed using React.js with Vite, offering real-time statistics including total emails received, SMS sent count, failed SMS count, and a complete SMS delivery log with SUCCESS and FAILED status indicators. The environment-based configuration system using a .env file ensures secure management of sensitive credentials including Gmail App Passwords and SMS API keys.

This project demonstrates the practical integration of networking protocols (IMAP, HTTP, REST), web technologies, and mobile communication to create a cost-effective, reliable real-time notification system applicable to students, IT professionals, and business users requiring offline SMS alerts for time-sensitive emails.

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